DRIVER MATCHING: CURRENT SYSTEM

How matching works now, how drivers are scored, and what is still open

Updated: October 25, 2026 | **Replaces:** the September 30 version by Akshay | **Sources:** the matching engine code (goezz-nodejs-server, branch UAD-1024, last commit Oct 21), Jira ticket UAD-1024 and its comments to Oct 21, the repo workflow note, and the live settings table (admin_meta_data) read on Oct 25.

Status: Jira marks UAD-1024 "Done (Ready for prod)" (Oct 23). I did not confirm that this branch is deployed to production, so check before quoting these rules as live.

1 Summary

- **The September problems are largely fixed.** The scoring rules that were switched off (route history, rider history, distance) are now on. A family's regular driver now outranks a closer stranger.
- **Matching no longer gives up.** It runs a round every 15 minutes, 8 new drivers per round, and the search widens by 2 miles per round up to 30 miles, until a driver claims the ride, or it is cancelled or started.
- **Rides are published later.** A ride is first offered 6 days before pickup, not 24 hours before.
- **Live location is now used** as an extra way to qualify and as a distance score.
- **Still open:** several QA bugs reported Oct 22 (section 7), and the final-round path has only been covered by automated tests, not a live run.

2 When matching runs

- **Daily job, 7:00 PM Pacific:** publishes every unclaimed ride group that has reached the 6-day mark (144 hours before pickup). Rides more than 6 days out are not published, and edits or new requests do not start matching for them.
- **Rides created inside the 6-day window** are published as soon as possible (at creation, or by the next 7 PM job).
- **Shared rides:** a ride is held for a possible shared match only if the parent chose "shared ride required", the pickup or drop-off is a designated school or activity location, and it is the only ride on its route. A held ride is published when another ride joins, when pickup is within 72 hours, or when the parent changes the preference. No locations are designated yet, so nothing is held today.
- **Last-minute rides** (inside 12 hours, the live setting) get their own single-ride route and are matched immediately.
- **An admin pressing Publish** always publishes at once.
- **Other triggers:** a driver declining or abandoning (a replacement is matched the same minute), a driver accepting a ride (conflicting offers are removed and replacements matched), a driver coming online with a fresh location (offered the open rides they qualify for), and a retry loop every 15 minutes while the ride is unclaimed.

3 How a driver is selected

For each ride group the engine runs these steps in order:

- **A. Exclusions.** Drivers already offered this ride, offered before, who declined or abandoned it, or who are blocked by the parent.
- **B. Hard filters (all must pass).** Onboarding passed and active; an active primary vehicle with enough seats; gender preference when the rider requires it (enforced, so it is a filter and gives no points); no claimed

ride at the same time; able to reach this ride from their previous ride and reach their next ride in time (Google Routes with live traffic plus a 5-minute buffer); and at least one ride group completed in the past 7 days. If nobody qualifies as active, other qualified drivers are considered so the ride is not left unmatched. A driver with an active match block (section 4) is also left out.

- **C. Proximity (straight-line miles).** A driver qualifies if any one is within the radius: their service location to the pickup; their live location to the pickup (only when the ride is at least 2 hours away and the location is under 2 hours old); their previous ride's end to the pickup; or their next ride's start to the drop-off. Home address is not used. A cancelled or empty ride group does not count as a nearby ride.
- **D. Rider-history exception.** A driver who completed a ride for the same rider in the past year is considered even outside the radius, up to 30 miles, but must still pass every hard filter.
- **E. Expansion.** Start at 4 miles. If fewer than 8 qualify, widen by 2 miles (up to 4 times: 6, 8, 10, 12). The radius is not carried into the next round.
- **F. Score, sort, offer.** Each driver is scored (section 4), ties are shuffled at random, and the top 8 new drivers get the offer and an app notification.

Rounds. Rounds 1 and 2 are the normal rounds. Round 3 is the "final" round and reaches 14 miles; each later round widens by another 2 miles (16, 18 and so on) up to a 30-mile cap. Every round offers the ride to 8 new drivers. A replacement after a decline, or a driver who just came online, is offered one driver at a time.

4 How driver scoring works

Every driver who passes the filters starts at 0 points. Points are added and subtracted by the rules below. The highest total is offered first; the top 8 get the offer. Each driver's point breakdown is saved in the match log that admins can see.

Rule	Points (live settings)	How it is counted
Rider history	+200 per rider	Each rider on this ride group that the driver completed a ride for in the past year.
Route history	+50 per location	Each pickup or drop-off location on this ride group that the driver completed a ride for in the past year.
Distance	+100 / +80 / +50 / +20	Closest of the driver's service location, live location, or nearby ride to the pickup. Within 3 miles = 100, within 5 = 80, within 8 = 50, within 12 = 20. The tightest bracket the driver fits is used. Beyond 12 miles = 0.
Completed rides	+10 to +50	Rides completed in the past 365 days, in bands: 10 rides = 10, 20 = 20, 30 = 30, 40 = 40, 50 or more = 50.
Recent activity	+2 per ride group	Ride groups the driver completed in the past 7 days.
Gender preference	+125 (not used today)	Only applies when the filter is switched off. Today it is a hard filter, so it gives no points.
Reliability penalty	minus active penalty points	Abandon = 30, quick abandon (within 15 minutes of accepting) = 5, last-minute abandon (within 12 hours of pickup) = 500. A penalty expires after 14 days.

Worked example. Driver A lives 2.5 miles from the pickup (100), has done 35 rides in the past year (30), has driven this rider before (200) and served one of the locations (50). Total 380. Driver B lives 2 miles away (100) and has done 60 rides (50), but has never driven this family. Total 150. Driver A is offered the ride first, even though B is closer. If A had one abandon penalty (30) the total would be 350, still ahead.

Match block. A driver with 2 or more active abandons (or any active no-show) is blocked from new matches until the penalty expires. The driver is not deactivated and keeps access to the app. The admin picker shows "Match Blocked until...".

Not used for ranking: home address, and the driver's earlier acceptance rate. Tiered exposure (showing the ride to the best few first, then widening) is not built.

5 Live settings (admin Meta Data)

Setting	Value	Meaning
driver-proximity-miles	4	Starting radius
driver-proximity-miles-match-increment	2	Miles added per expansion
max-expansion-attempts	4	Expansions per round
driver-proximity-max-miles	12	Radius cap for normal rounds
driver-match-continued-max-miles	30	Radius cap once matching continues
number-drivers-to-match	8	New drivers per round
max-match-attempts	2	Normal rounds before the final round
match-interval	900 seconds	Wait between rounds (15 minutes)
driver-match-continue-until-claimed	1 (on)	Keep going until claimed, cancelled or started
last-minute-threshold-minutes	720	Last-minute window (12 hours)
feasible-travel-buffer-minutes	5	Extra time allowed for travel between rides
prior / after-ride-group-qualifying-minutes	40 / 40	Window for a nearby previous or next ride
driver-match-active-drivers-only / lookback	1 / 7 days	Only drivers active in the past 7 days
ride-group-publish-lead-hours	144	Publish 6 days before pickup
shared-ride-hold-cutoff-hours	72	Held shared rides publish at this point

One thing to confirm: the table also holds an older combined setting, "driver-match-engine-params", with different numbers (for example distance points 500 / 400 / 100, rider points 50, 3 attempts). The code on this branch does not read it, so it appears to be unused. Remove or label it so nobody edits the wrong one. I also cannot tell from the database connection alone whether it points at production or a test copy, so confirm the values in the admin screen.

6 Status of the problems found in September

Problem (Sept 30)	Status now
1. The regular driver is not prioritised	Fixed. Route history (50) and rider history (200) scoring is on, and a driver who served the rider is considered out to 30 miles. Retested Oct 21.
2. Being closer gives no advantage	Fixed. Distance points count (for example 100 within 3 miles). Retested Oct 19.
3. "Recently completed" scored zero	Fixed. The setting is read; live value is 2 points per ride group in the past 7 days.
4. Service address missing for some drivers	Mitigated. The drivers list shows "Missing - cannot be matched" and a weekly Monday 9 AM text reminds those drivers. Not yet required at onboarding.

Problem (Sept 30)	Status now
5. Live location not used	Done. Used as a qualifier and for distance points when the ride is at least 2 hours away; also triggers offers when a driver comes online. Cannot be tested from outside Seattle.
6. Driver with no primary contact dropped silently	Not re-checked in this review.
7. Smaller code issues	Mostly fixed: every round now offers 8 drivers; the radius widens one step per round; the Match Engine test page is a read-only dry run.

7 What is still open

- Admin can match a driver to an overlapping route with no warning, and the picker does not label busy drivers (QA, Oct 22).
- After a cancel empties a route, the freed driver was not considered for a new ride in the same slot in two of three tests.
- Routes created Oct 19 before the fix stopped after 2 to 3 attempts. They need to be republished to keep matching.
- A last-minute sibling cancel does not make the driver re-acknowledge the route, though the price changes.
- The decline dialog still mentions "account inactivation" instead of the match-block wording.
- Nearby-ride proximity (a driver qualifying only because of a ride next to this one) is only partly confirmed.
- The final-round path (up to 30 miles) was covered by automated tests only. Watch the first real final round.
- The shared-ride list of designated locations is empty, and the parent apps do not yet send "shared ride required".
- Point weights and the 72-hour hold still need business sign-off before they are treated as final.

Prepared from the code and tickets listed above. Values are the current settings and may be tuned from the admin Meta Data screen. Original author of the September version: Akshay Mistry.